Simulation-Based Inference

Exercise 5

NOTE: *The solution of this exercise will be discussed in the tutorial session on 08.06.2026 along with next week's exercises.*

Exercise 5.1: Simple Markov Chains

a) Approximate the stable distribution of a Markov Chain on a discrete space with three states $(1, 2, 3)$, initial distribution $p_0 = (0.1, 0.7, 0.2)$ and transition matrix

$$Q = \begin{pmatrix} 0.4 & 0 & 0.6 \\ 0.3 & 0.2 & 0.5 \\ 0.1 & 0.1 & 0.8 \end{pmatrix}.$$

Make sure to use a sufficiently long warmup phase and to obtain at least 1000 post-warmup samples.

b) Suppose you would have chosen the initial distribution differently as $p_0 = (0.1, 0.2, 0.7)$. Now that you have seen the stable distribution in (a), would you expect slower or faster convergence with the new initial distribution? Justify your answer.

Exercise 5.2: Metropolis Algorithm

Consider a simple Bayesian model where our goal is the estimate the mean of a total of N conditionally independent observations $y_1, \ldots, y_N$.

We assume a normal likelihood

$$y_i \sim \text{normal}(\theta, 10) = p(y_i \mid \theta)$$

and a flat prior

$$p(\theta) = \text{const}$$

As proposal distribution, we choose:

$$\theta^{(p)} \sim \text{normal}(\theta^{(t)}, 0.5)$$

This means that we can write our posterior density ratio α as

$$\alpha = \frac{p(\theta^{(p)} \mid y)}{p(\theta^{(t)} \mid y)} = \frac{\prod_{i=1}^{N} p(y_i \mid \theta^{(p)})}{\prod_{i=1}^{N} p(y_i \mid \theta^{(t)})}.$$

On the log scale, this can be written as follows:

$$\log \alpha = \log p(\theta^{(p)} \mid y) - \log p(\theta^{(t)} \mid y) = \sum_{i=1}^{N} \log p(y_i \mid \theta^{(p)}) - \sum_{i=1}^{N} \log p(y_i \mid \theta^{(t)}).$$

a) Implement the metropolis algorithm for data y and starting value $\theta^{(1)}$.

b) Simulate a random dataset $\tilde{y}$ of $N = 200$ observations from the above likelihood with a true parameter value of $\theta^* = 2$. Compute the posterior $p(\theta \mid \tilde{y})$ with your metropolis implementation using 4 independent chains initialized from different starting values.

c) Check whether your posterior samples from (b) have converged.

d) Visualize the posterior samples and compare the obtained posterior with the ground truth parameter value $\theta^* = 2$.